

ABSTRACT

The Aerospace field relies heavily on structures that have excellent mechanical and thermal properties with fatigue resistance and high fracture toughness. This leads to a significant use of Anisotropic Materials including alloys and composites in the manufacturing of Aircrafts and Satellites.

Anisotropic Materials have physical properties that depend on the direction of measurement for e.g., Young's Modulus, yield strength, creep rate, etc. Carbon Fiber-Reinforced Polymers (CFRP) are used in wings and fuselage due to their high strength to weight ratio, Kevlar has high tensile strength and is therefore used in space shuttles to protect against impact from orbital debris. Aluminium alloys are also used extensively in aeroplanes because of their high strength-to-weight ratio, corrosion resistance and low cost.

Since anisotropy is complex and the properties are crystal structure dependent, Crystal Plasticity Theory is used to understand and determine the micro-scale deformation of metals. According to Crystal plasticity theory, deformation occurs through the motion of dislocation in a slip system. This theory also provides the stress-strain relationship of the grains in a crystal.

However, determination of anisotropic properties through crystal plasticity theory requires high computational costs and effort. To combat this problem, this paper aims at determining the properties of a AA5083 polycrystal by feeding information obtained from crystal plasticity simulations in DAMASK to a Convolutional Neural Network. The ability of neural networks to understand complex relationships between features and predict the desired output with high accuracy would contribute significantly in reducing the challenges associated with crystal plasticity simulations.

The simulations were carried out by Lukas Martinitiz, Researcher at TUM and my supervisor. The CNN model was trained to predict stresses and new orientations produced due to deformation gradients applied on the Representative Volume Element (RVE), while also preserving the relative position of grains. The model was evaluated by calculating the Root Mean Squared Error (RMSE) for stresses and Mean Absolute Error (MAE) for orientations and the results were discussed. The CNN replicated the Crystal Plasticity simulation information with 2% training and validation loss and 31 MPa stress RMSE and 16% orientation MAE.